

1.11.2 Testing Methods (Usability, A/B, Automated)

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CS XII

Chapter 1: Computer Systems (HCI)

Lecture 15 of 16

★ Golden

1.11.2 Testing Methods (Usability, A/B, Automated) ★ Golden

LEARN IT

Testing involves observing how the system behaves under specific conditions. Learn all seven methods. The three named in the heading — usability, A/B and automated — are the ones examiners ask most, but a complete answer also names the others.

1. **Usability testing:** observing real users while they perform realistic tasks, to measure efficiency and satisfaction. If the user gets stuck, designers know where to fix the UI.
2. **Task-based testing:** assigning a specific task (for example, "Find a product and add it to cart") and strictly measuring **completion time**, **success rate** and **number of errors**.
3. **Automated audits:** software bots scan websites to detect technical issues such as broken links, slow loading speeds or low colour contrast.
4. **A/B testing:** comparing two versions of an interface (A and B). Half the users see A, half see B. The one with more clicks or sign-ups wins.
5. **Compatibility testing:** verifying the system on different browsers (Chrome, Safari), devices (mobiles, tablets, PCs) and operating systems.
6. **Performance and load testing:** simulating high traffic (for example 10,000 users logging in at once) to see if speed drops or the server crashes.
7. **Keyboard and screen-reader walkthroughs:** navigating using *only* the keyboard (Tab/Enter) or assistive technologies, to check accessibility for users with disabilities.

Task-based "Add to cart" Time: 12 s Success: Yes	A/B test Version A Version B half / half users more clicks wins	Automated audit Bot scanning... 3 broken links Low contrast
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Three golden testing pictures: timed tasks, A versus B, and a bot that finds broken links and weak contrast.

WORKED EXAMPLE

Worked example: test JazzCash "Send Money" and a school portal with numbers.

Usability testing. Sit beside five students. Task: "Send Rs 100 to this test number." Do not help. Note where they pause. If three people look for Confirm at the bottom and it is at the top, that is your UI fix.

Task-based testing (numbers). Same task for 10 students.

- Success: 8 of 10 finish = success rate $(8 \div 10) \times 100 = 80\%$.
- Times: 12, 14, 11, 40, 13, 12, 15, 50, 14, 12 seconds. The two slow times (40, 50) are the students who got stuck — they also made errors.
- Errors: extra taps, wrong field. Count them. High errors + high time = poor HCI, even if most people eventually succeed.

A/B testing. Version A: green "Send" at the top. Version B: large orange "Send" under the amount. 200 users each. If B gets 140 completed payments and A gets 90, B wins. You keep B.

Automated audit. A bot reports 3 broken "Fee challan" links and low contrast on the portal's grey-on-grey marks table. Fix those before the board-exam rush.

Load testing. On result day, 10,000 students log in at 9:00 a.m. If the server crashes, performance testing failed — the UI never even appears.

Keyboard walkthrough. Unplug the mouse. Can a student Tab to "Download date sheet" and press Enter? If focus jumps randomly, accessibility testing failed.

REMEMBER IT

Task-based three numbers: Time, Success, Errors. **A/B:** two versions, split users, the higher clicks/sign-ups win. **Bot:** links, speed, contrast. **Load:** many users at once. **Keyboard:** Tab and Enter only.

COMMON MISTAKE

Mistake: Calling A/B testing "showing both designs to the same user and asking which is prettier".

Fix: Half the users see only A, half see only B. You measure behaviour (clicks, sign-ups), not a taste vote. Also: an automated audit does not replace watching real users — bots cannot see that a student is confused.

اردو میں سمجھیں

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EXAM TIP

Golden 8-mark question: "Explain testing methods used in HCI." Write seven short paragraphs or a seven-row table. For full marks, define usability testing (observe real users on realistic tasks), give the three task-based measures, describe automated audits with the three bot findings (broken links, slow load, low contrast), explain A/B with the half-and-half rule, then add compatibility, load (10,000 users), and keyboard/screen-reader walkthroughs. If the paper asks only "usability, A/B and automated", go deep on those three and still mention the rest in one closing line.

CHECK YOURSELF

1. Name the three measurements in task-based testing.
2. In A/B testing, who sees version A?
3. Give two issues an automated audit can detect.

Answers: 1. Completion time, success rate, number of errors. 2. Half of the users (the other half see B). 3. Any two of: broken links, slow loading, low colour contrast.

Next lecture: Career Opportunities in UI/UX